

CAD CAM Database

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- Computer-aided design can increase the productivity of the design department in the company
- CAD system offers the opportunity to develop the data base needed to manufacture the product
- In the conventional manufacturing cycle - *engineering drawings were prepared by design draftsmen* and then used by *manufacturing engineers to develop the process plan*
- The activities involved in designing the product were separated from the activities associated with process planning.
- The two-step procedure was employed was time consuming and involved duplication of effort by design and manufacturing persons.
- In an *integrated CAD/CAM system, a direct link is established between product design and manufacturing.*

CAD CAM Database

It includes all the data on the product generated during design (geometry data, bill of materials and parts lists, material specifications, etc.) integrated with data required for manufacturing

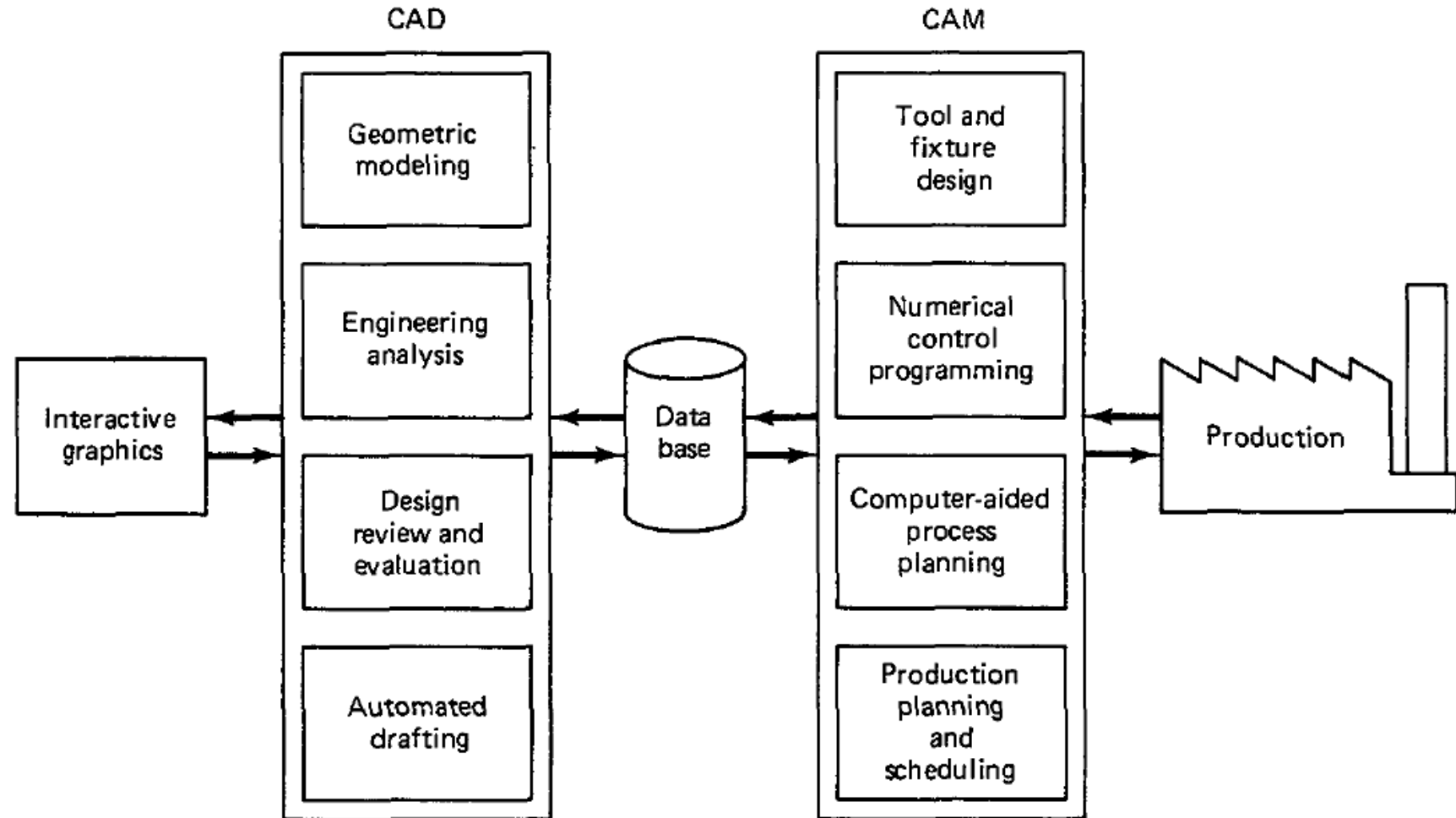


FIGURE 4.10 Desirable relationship of CAD/CAM data base to CAD and CAM.

Benefits of CAD

1. Improved engineering productivity
2. Shorter lead times
3. Reduced engineering personnel requirements
4. Customer modifications are easier to make
5. Faster response to requests for quotations
6. Avoidance of subcontracting to meet schedules
7. Minimized transcription errors
8. Improved accuracy of design
9. In analysis, easier recognition of component interactions
10. Provides better functional analysis by prototype testing
11. Assistance in preparation of documentation
12. Designs have more standardization
13. Better designs provided
14. Improved productivity in tool design
15. Better knowledge of costs provided
16. Reduced training time for routine drafting tasks and NC part programming
17. Fewer errors in NC part programming
18. Provides the potential for using more existing parts and tooling
19. Helps ensure designs are appropriate to existing manufacturing techniques
20. Saves materials and machining time by optimization algorithms
21. Provides operational results on the status of work in progress
22. Makes the management of design personnel on projects more effective
23. Assistance in inspection of complicated parts
24. Better communication interfaces and greater understanding among engineers, designers, drafters, management, and different project groups

CAD Data Exchange

- Use CAD model data for various CAD based engineering applications like production drawings, 3Dprinting, FEA, CFD, CAE, CAM etc.
- Geometric data exchange refers to conversion from one geometric data format to another.
- CAD/CAM software developers use different formats to store the data.
- *Fundamental incompatibilities among entity representations further complicate exchanging the modelling data.*
- The Geometry data format in which part geometry information is stored often varies from one CAD/CAM system to another and hence the *format need not be unique one*

Need for geometric data exchange

Heterogenous expertise in industry

- In one organization expertise is available in one CAD package while in other organization people are more familiar with other package.
- This heterogeneity will always exist as long as number of alternative packages are available in market.
- That arises the need of data exchange.

Use of application specific packages

- Single package may not satisfy all the requirements of design and manufacturing.
- Even if only one versatile comprehensive package is used for all CAD/CAM operations, still data exchange is needed from one module to another.

Need for geometric data exchange contd.

- Migration from one system to another
 - May be because of availability of better package in the market.
 - Lack of technical support from the parent company.
- Data exchange with collaborators/ customers/suppliers
 - Because of technology transfer
 - Data exchange between developer and beneficiary
- Rapid pace of technological change

Data that needs to be exchanged

Shape data

- Geometrical Information
- Topological Information
- Attributes such as font, colour, material
- Annotations

Non-shape data

- It includes graphics data such as shaded images
- Measuring units
- Resolution

Data that needs to be exchanged contd.

Design data

- Information that designers generate from geometric models for analysis purpose
- Mass property and finite element mesh data belong to this type of data.

Manufacturing data

- It consists of information as tooling, NC tool path,
- Tolerancing, process planning, tool design and bill of materials

Options available for data exchange

- Native format
- Neutral format
- Standard format
- Binary format
- ASCII format

Options available for data exchange

Native Format:

- This is Company's own format
- Generally not disclosed
- Always with some specific extension.
- File can be read only by the same package.
- It is not a standard.
- It is not opened by another package unless there is a translator.

Options available for data exchange

Neutral format:

- package gives an optional format which can be very easily interpreted by the outside world.
- Generally it comes with all information like what type of data is stored , what is the order of information etc...
- It may come with manual wherein everything is mentioned about how data is stored.

Standard format:

- Accepted by national or international standard.

Options available for data exchange

Binary format

A binary file is a computer file that is not a text file.

ASCII format

- *American Standard Code for Information Interchange*
- It is a standard encoding for alphanumeric characters used in computers and related devices.
- ASCII was introduced by the United States of America Standards Institute (USASI) now known as the American National Standards Institute.

ASCII - Binary Character Table

Letter	ASCII Code	Binary
A	065	01000001
B	066	01000010
C	067	01000011
D	068	01000100

Different formats

Some standard available formats are-

- IGES, DXF, STL, STEP, PDES, ACIS, Parasolid
- DXF- Drawing Exchange Format
- IGES- Initial Graphics Exchange Format
- STL- Standard Triangular Language
- STEP- Standard for Exchange of Products data

IGES (Initial Graphics Exchange Specification) format

- It is the first standard exchange format
- IGES has been revised a few times since its version 1.0 was released in 1980
- The various IGES versions share common characteristics
- Each version must remain upwards compatible with previous version
- This means that a processor that is fully conforming to the latest version can correctly interpret IGES files written in accordance with prior versions

IGES (Initial Graphics Exchange Specification) format

- CAD/CAM vendors or companies specialized in database transfer must write software to translate from their systems to IGES format and vice versa.
- The software that translates from the native database format of a given CAD/CAM system to the IGES format is called a *preprocessor*
- The software that translates in a opposite way (from IGES to a CAD CAM system) is called as *postprocessor*
- The preprocessors and postprocessors are also called as translators